MNLP: Homework 1

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1 Introduction

This report evaluates transformer-based embeddings for semantic retrieval. We study training strategies such as sentence pairs, triplets, and hard negatives, along with extensions including cross-encoder re-ranking and re-chunking.

2 Methodology

2.1 Pair, Triplet, and Hard Negative Training

We initially explore standard pairwise and triplet objectives to establish a baseline embedding space. We further improve training by incorporating **hard negatives**, defined as candidates that are semantically close to the query but incorrect. This approach teaches the model to distinguish between subtle differences in candidate chunks.

2.2 Re-ranking

After retrieving the top-k candidates using the bi-encoder, we re-score these candidates using a cross-encoder model.

2.3 Re-chunking

We also experimented with alternative chunking strategies. Instead of using the original dataset chunks, we explored re-splitting the text into smaller segments to reduce noise and irrelevant tokens within each correct answer. For this task, we treat the original chunks as the ground truth, assuming they contain the relevant context needed to answer the query.

2.4 Evaluation metrics

We evaluate using Hit@k and MRR, reporting cosine similarity as the primary metric and comparing it with Euclidean distance without normalization, since under the L_2 norm both metrics are equivalent. After fine-tuning, we additionally evaluate chunk token usage and answer retention.

3 Experimental setup

To maintain experimental consistency two pipelines were implemented: one for embedding generation and ranking, and one for fine-tuning. All experiments were conducted with a 7:1 train-validation split of the training data.

3.1 Model Choices

All experiments are performed on `distilbert-base-uncased` [2], `all-MiniLM-L6-v2` [1], and `BAAI/bge-base-en-v1.5` [3]. The BGE model was chosen since it has been trained for retrieval tasks and offers a good size-to-performance.

3.2 Loss functions

We use two main loss functions during fine-tuning. For sentence pairs and hard negative training, we use *MultipleNegativesRanking (MNR) loss*, where other samples in the batch act as negatives for the current query. This allows the model to efficiently learn from a large number of implicit negatives. For triplet training, we use a standard triplet loss to explicitly enforce relative ordering between the correct chunk and a sampled negative, using a margin of 0.5.

3.3 Fine-tuning

We first experimented on the Distilbert model with different fine-tuning techniques, then applied the most effective way to the other models to compare performance. Due to GPU memory constraints, batch sizes varied across experiments, which affects the number of implicit negatives when using MNR loss. We used early stopping with patience 2 based on validation performance.

3.3.1 Sentence Pairs and Triplets

We created a dataset of (Query, correct_chunk) pairs, and one of (Query, correct_chunk, random_chunk) triplets to use during fine-tuning. The random chunk is, arbitrarily, the chunk immediately following the correct chunk.

3.3.2 Hard Negatives

These samples are obtained through the predictions of the baseline models. For each query, we sample the 9 most similar incorrect chunks together with the correct one. Using *MultipleNegativesRanking Loss*, each batch implicitly provides additional negatives, since all other positives in the batch act as negatives for the current query.

3.4 Re-ranking Setup

For re-ranking, we first retrieve the top-k candidates using the trained bi-encoder. These candidates are then passed to a cross-encoder, which produces a more refined ranking score. For the cross-encoder, we experimented with `BAAI/bge-reranker-base` [3] and `cross-encoder/ms-marco-MiniLM-L-12-v2` [1].

3.5 Re-chunking Setup

Since a major real-world issue is token consumption, our re-chunking approach focuses on isolating the answer to the query from the selected chunk. First, the bi-encoder retrieves the predicted chunk, then we split it into single sentences. These sentences are passed through the final attention layers of the BGE cross-encoder, allowing us to observe how the query attends to different candidate sentences. After that, the sentence with the highest attention score is selected, along with one adjacent sentence, to maintain coherence. Table 8 illustrates this.

This doesn't improve retrieval quality, instead, it isolates the correct information post-hoc.

4 Results

A comprehensive table of results can be found in Table 6 in the appendix.

4.1 Baseline

The baseline results of `distilbert-base-uncased` and `all-MiniLM-L6-v2` correspond with the reference results.

4.2 Fine-tuned

Figures 1 and 2 show that all experiments started overfitting after 1-3 epochs.

Sentence Pairs. The only model fine-tuned using this approach was `distilbert-base-uncased`. The Hit@1 score increased from 0.17 to 0.32 and MRR increased from 0.36 to 0.50.

Sentence Triplets. We further fine-tuned the DistilBERT model on sentence triplets. This increased the Hit@1 score to 0.52, and the MRR to 0.65. While this improves over using sentence pairs, it still does not capture semantic nuances as effectively as hard negative training.

Hard Negatives. Hard negative training was the most effective fine-tuning strategy. Instead of using a shared negative pool, we mined hard negatives separately for each model using its own retrieval predictions. This makes the negative set adaptive to the model's current failure modes.

This improved DistilBERT's Hit@1 from 0.17 to 0.61 and MiniLM's from 0.44 to 0.62. Fine-tuning BGE on hard negatives resulted in our best model. Its Hit@1 increased from 0.54 to 0.68, and it achieved an MRR of 0.79.

This is likely due to BGE's retrieval-oriented pretraining and instruction-style query formulation, which align well with the task.

4.3 Re-ranking

We added an off-the-shelf cross-encoder re-ranking step on top of the bi-encoder retriever. However, this degraded performance.

A possible explanation is that the cross-encoder was not well aligned with the task-specific distribution of the dataset, leading to inconsistent scoring. While this could have been mitigated by fine-tuning the cross-encoder, this was precluded due to resource constraints.

4.4 Re-chunking

A separate cross-encoder (`ms-marco-MiniLM-L-6-v2`) was used to assess whether the selected subchunks contained the correct answer. Since this model outputs raw logits, a softmax was applied to obtain probabilities. A prediction was considered correct if its probability exceeded 50%, or if it remained within a 10% margin of the original chunk's score when both were below this threshold.

Under this evaluation, the average context size was reduced by 44.1%, while correct answers were retained in 91.5% of cases. Table 7 summarizes the results. This suggests that the approach can be effective at preserving relevant information while significantly reducing input length.

References

[1] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.

[2] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *ArXiv*, abs/1910.01108.

[3] Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. 2023. C-pack: Packaged resources to advance general chinese embedding. *Preprint*, arXiv:2309.07597.

A Reproducibility Setup

All the trainings were done using Nvidia L4 24GB GPU. Batch sizes are adjusted accordingly for each model depending on its parameter size. We fixed the random seed to 42 for all libraries.

B Hyperparameters

Our models were trained using the AdamW optimizer. Detailed hyperparameters for the final reported runs are provided in Tables 1- 5.

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	$2e-5$
Batch Size	32
Loss Function	MNR
Training Epochs	20 (Early stopping)
Patience	2

Table 1: Hyperparameters used for DB on SP

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	$2e-5$
Batch Size	32
Loss Function	Triplets
Training Epochs	10 (Early stopping)
Patience	2

Table 2: Hyperparameters used for DB on ST

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	$2e-5$
Batch Size	16
Loss Function	MNR
Training Epochs	15 (Early stopping)
Patience	2

Table 3: Hyperparameters used for DB on HN

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	$3e-6$
Batch Size	8
Loss Function	MNR
Training Epochs	15 (Early stopping)
Patience	2

Table 4: Hyperparameters used for AM on HN

C Fine-Tuning Results

Table 6 reports the complete test-set results for all model and training variants. The main trend is consistent across models: hard negative training gives the largest improvement, especially for top-ranked retrieval.

DistilBERT benefits the most from fine-tuning because it starts as a general-purpose encoder. Pair training improves Hit@1 from 0.1765 to 0.3265, triplet training further improves it to 0.5295, and hard negative training reaches 0.6105. This shows that moving from simple alignment to harder ranking guidance progressively improves the embedding space.

MiniLM starts from a much stronger baseline due to sentence-similarity pretraining, but still improves with hard negatives, reaching 0.6240 Hit@1. BGE gives the best overall results, with hard negative training reaching 0.6845 Hit@1 and 0.7941 MRR. This is likely due to its retrieval-oriented pretraining, instruction-guided query encodings and larger parameter size (109M parameters vs. 66M for DistilBERT and 22M for MiniLM), which allows it to better capture semantic distinctions.

The Euclidean results are generally close to cosine for DistilBERT, but cosine remains the primary metric because it is standard for normalized dense retrieval. The Euclidean and cosine results are identical for MiniLM and BGE. This is because both models produce normalized embeddings, making Euclidean distance and cosine similarity equivalent under the L_2 norm. For DistilBERT, embeddings are not strictly normalized, leading to small differences between the two metrics.

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	$5e-6$
Batch Size	4
Loss Function	MNR
Training Epochs	10 (Early stopping)
Patience	2

Table 5: Hyperparameters used for BGE

D Fine-Tuning Loss Curves

Figures 1 and 2 show consistent behavior across all models and training strategies. In all cases, training loss decreases steadily, while validation loss improves only in the first few epochs and then plateaus or increases, indicating early overfitting.

DistilBERT exhibits the largest gap between training and validation loss, suggesting weaker generalization and higher sensitivity to overfitting. MiniLM shows more stable behavior, with a smaller gap and smoother validation trends. BGE converges the fastest, reaching its best validation performance early, after which overfitting becomes evident.

Overall, the curves confirm that most of the performance gains occur in the early epochs, and that early stopping is critical, especially for larger or retrieval-specialized models.

E Re-Chunking Details

E.1 Attention-Based Sentence Selection

We first use the bi-encoder to retrieve the most relevant chunk for a given query. This chunk is then split into individual sentences. To identify the most relevant sentence, we leverage the attention mechanism of a cross-encoder (`bge-reranker-base`).

The BGE cross-encoder consists of 12 transformer layers. We extract the attention weights from the final layer, which contains 8 attention heads, and average them to obtain a single attention map. This produces token-level interactions between the query and the chunk.

From this map, we compute a sentence-level relevance score by aggregating attention values between query tokens and tokens in each sentence. We experimented with both average and sum aggregation. However, average attention introduces a bias toward shorter sentences, which can artificially inflate their scores. To avoid this, we use the sum of attention values as the final scoring function.

E.2 Design Choice and Evaluation

We retain the top-ranked sentence together with one adjacent sentence, resulting in a maximum of two sentences per chunk. This is not an arbitrary choice. Upon inspecting many examples, we observed that most queries target a single piece of information, which is typically contained within one or two sentences. This makes the approach both efficient and sufficient in practice.

To evaluate whether the reduced chunks still contain the correct answer, we use a separate cross-encoder (`ms-marco-MiniLM-L-6-v2`). This avoids bias from the model used during selection. The model outputs logits, which we convert to probabilities via softmax. A reduced chunk is considered valid if its probability exceeds 50%, or if it falls within a 10% margin of the original chunk's score, indicating that the relevant information has been preserved.

The results in Table 7 show that this approach reduces context length by 44.1% while retaining correct answers in 91.5% of cases.

Token Reduction Example Table 8 illustrates qualitative evaluation of how our re-chunking strategy isolates relevant information from a selected chunk.

		Cosine				Euclidean			
Model	Method	Hit@1	Hit@3	Hit@5	MRR	Hit@1	Hit@3	Hit@5	MRR
DistilBERT	Baseline	0.1765	0.4435	0.6045	0.36	0.1650	0.4150	0.5810	0.3488
DistilBERT	Pair	0.3265	0.6180	0.7440	0.50	0.3045	0.5880	0.7190	0.48
DistilBERT	Triplet	0.5295	0.7465	0.8210	0.66	0.5245	0.7510	0.8270	0.65
DistilBERT	Hard Neg	0.6105	0.8270	0.8970	0.73	0.5935	0.8110	0.8895	0.71
MiniLM	Baseline	0.4480	0.7165	0.8080	0.60	...	...	...	...
MiniLM	Hard Neg	0.6240	0.8490	0.9220	0.75	...	...	...	...
BGE	Instr. Baseline	0.5460	0.8040	0.8815	0.69	...	...	...	...
BGE	**Hard Neg (Instr.)**	**0.6845**	**0.8860**	**0.9365**	**0.79**	...	...	...	...
BGE	Reranker	0.5650	0.8020	0.9015	0.70	...	...	...	...

Table 6: Full evaluation results on the test set using cosine similarity and Euclidean distance.

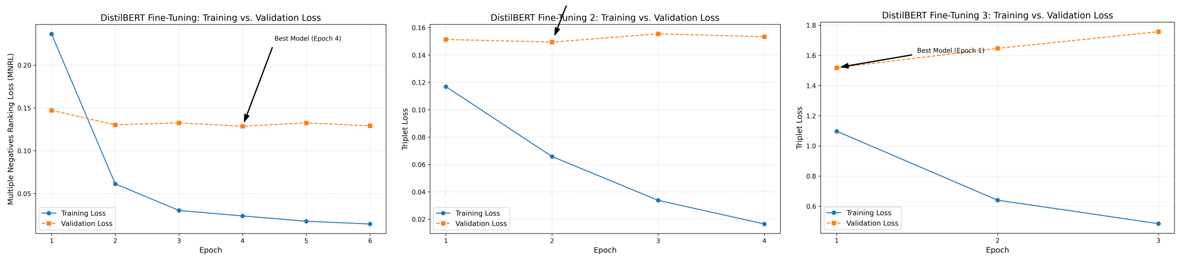

Figure 1: DistilBERT loss curves across the three fine-tuning stages: sentence pairs, triplets, and hard negatives.

Metric	**Value**
Average Original Chunk Size	116.1 tokens
Average Dynamic Chunk Size	64.9 tokens
Context Reduction	44.1%
Valid Baseline Answers	1805 / 2000
Answer Retention Rate	91.5%

Table 7: Effect of re-chunking on context size and answer retention on test set.

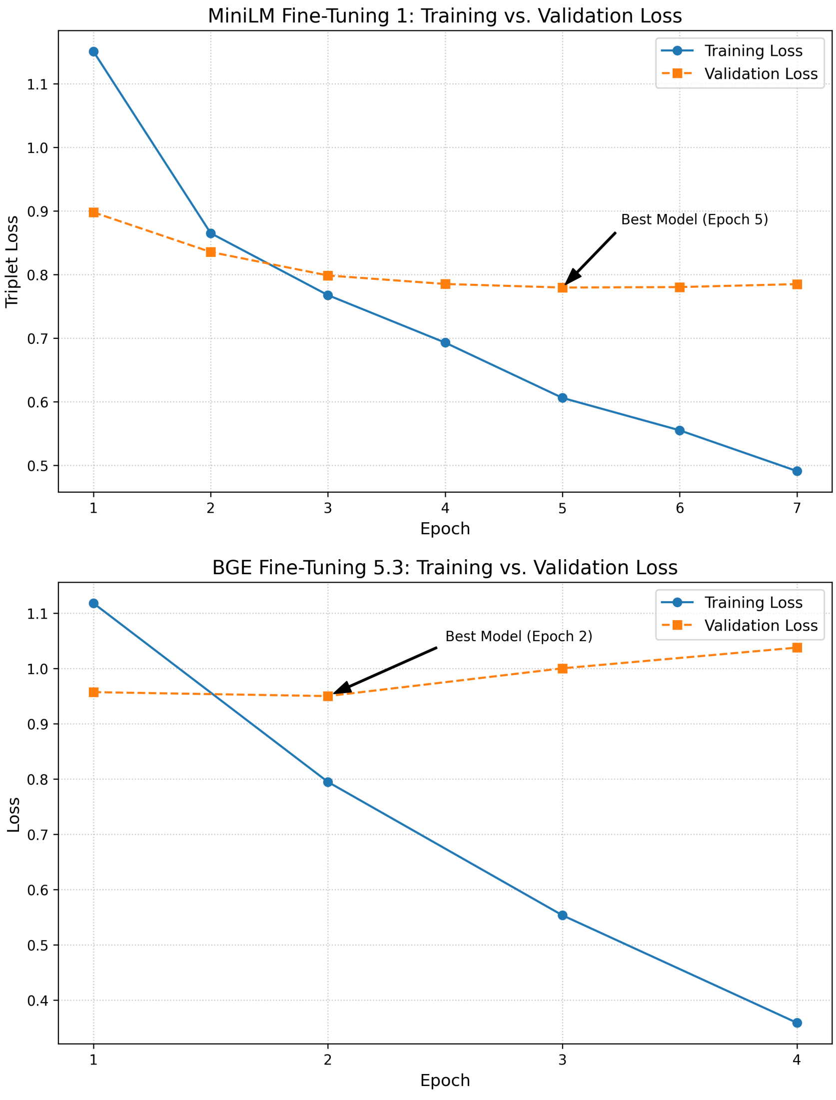

Figure 2: Hard negative training curves for MiniLM and BGE.

Query	Original Chunk	Dynamic Reduced Chunk
what is a woman's work lyrics about	The lyric of "This Woman's Work" is about being forced to confront an unexpected and frightening crisis during the normal event of childbirth. Written for the movie She's Having a Baby, director John Hughes used the song during the film's dramatic climax, when Jake (Kevin Bacon) learns that the lives of his wife, Kristy (Elizabeth McGovern), and their unborn child are in danger. As the song plays, we see a montage sequence of flashbacks showing the couple in happier times, intercut with shots of him waiting for news of Kristy and their baby's condition. Bush wrote the song specifically for the sequence, writing from a man's (Jake's) viewpoint and matching the words to the visuals which had already been filmed.	The lyric of "This Woman's Work" is about being forced to confront an unexpected and frightening crisis during the normal event of childbirth. Written for the movie She's Having a Baby, director John Hughes used the song during the film's dramatic climax, when Jake (Kevin Bacon) learns that the lives of his wife, Kristy (Elizabeth McGovern), and their unborn child are in danger.
who dies in the plane crash grey's anatomy season 8	"Flight" is the twenty-fourth and final episode of the eighth season of the American television medical drama Grey's Anatomy, and the show's 172nd episode overall. It was written by series creator Shonda Rhimes, and directed by Rob Corn. The episode was originally broadcast on the American Broadcasting Company (ABC) in the United States on May 17, 2012. In the episode, six doctors from Seattle Grace Mercy West Hospital who are victims of an aviation accident fight to stay alive, but Dr. Lexie Grey (Chyler Leigh) ultimately dies. Other storylines occur in Seattle where Dr. Richard Webber (James Pickens, Jr.) plans his annual dinner for the departing residents, Dr. Owen Hunt (Kevin McKidd) fires Dr. Teddy Altman (Kim Raver), and Dr. Miranda Bailey (Chandra Wilson) gets engaged.	"Flight" is the twenty-fourth and final episode of the eighth season of the American television medical drama Grey's Anatomy, and the show's 172nd episode overall. It was written by series creator Shonda Rhimes, and directed by Rob Corn.
where is the menu bar on apple computer	The Apple menu is a drop-down menu that is on left side of the menu bar in the classic Mac OS, macOS and A/UX operating systems. The Apple menu's role has changed throughout the history of Apple Inc.'s operating systems, but the menu has always featured a version of the Apple logo.	The Apple menu is a drop-down menu that is on left side of the menu bar in the classic Mac OS, macOS and A/UX operating systems. The Apple menu's role has changed throughout the history of Apple Inc.'s operating systems, but the menu has always featured a version of the Apple logo.
where do they live on malcolm in the middle	Much of the filming for Malcolm in the Middle was done on location in various parts of the Thirty Mile Zone around Los Angeles. A privately owned home, located at 12334 Cantura Street in Toluca Lake, California, was rented for upwards of $3,000 a day to film as Malcolm's house. Rebuilt in 2011, the property is no longer recognizable due to its modern two-floor design. However, the house directly to the left of it is nearly identical to what it looked like during filming, still making it a frequent stop for fans of the show. School scenes were filmed at Walter Reed Middle School, in North Hollywood, and the Lucky Aide was represented by a Drug Emporium at 6020 Lankershim Boulevard in North Hollywood. In "Stock Car Races," when Hal and the boys are entering a race track, the billboard behind the entrance displays the place as Irwindale Speedway, a real race track in Southern California. The last episode in the first season ("Waterpark") was filmed at a water park called Wild Rivers located in Irvine, California. It is never revealed which state the show is set in (except for Francis' whereabouts in early seasons, such as his military school in Alabama and his job in Alaska).	Much of the filming for Malcolm in the Middle was done on location in various parts of the Thirty Mile Zone around Los Angeles. A privately owned home, located at 12334 Cantura Street in Toluca Lake, California, was rented for upwards of $3,000 a day to film as Malcolm's house.
when did god of war remastered come out	God of War III Remastered is a remastered port of God of War III for the PlayStation 4 console. It was first released in North America on July 14, 2015, followed by Australia and mainland Europe on July 15, and the UK on July 17. Santa Monica's Creative Director Cory Barlog announced the remastered game in celebration of the God of War franchise's tenth anniversary. Ported by Wholesale Algorithms, the remastered version has full 1080p support targeted at 60 frames per second and features a photo mode, allowing players to edit their photos and share their favorite moments. All of the DLC that was released for God of War III is included with God of War III Remastered. By the end of its first week of release, God of War III Remastered was ninth in sales at retail in the UK. For the entire month of July 2015, the downloadable version was the seventh best-selling PlayStation 4 title from the PlayStation Store.	God of War III Remastered is a remastered port of God of War III for the PlayStation 4 console. It was first released in North America on July 14, 2015, followed by Australia and mainland Europe on July 15, and the UK on July 17.

Table 8: Full qualitative comparison between original and dynamically re-chunked outputs.